

ColliderAgent revision: audit of open placeholders

measurements of 2026-09-16/17 and proposed text

2026-09-17

Legend

DONE = measured today, text below can be pasted. **PARTIAL** = partly measured; the rest is in progress. **RUNS** = needs the remaining benchmark runs (Opus 4.6 $\times$ 3 per benchmark; cross-model). **AUTHOR** = a decision or confirmation only the authors can make.

1 Table S2 (sm.pdf p.4) — software environment

Status: **DONE** for every package (all read inside the images); only the Claude Code version of the *original* runs is open (see note). All versions were read inside the images on zhustation on 2026-09-16 with a probe blueprint (**collider-env-probe**; job logs are cited per row). Two corrections to the current table: the analysis Python stack lives in the *agent runtime*, not in the collider image (the image has no **uproot**), and Delphes is 3.5.1.

Image / runtime	Package	Version
collider	MadGraph5_aMC@NLO	3.7.0
collider	Pythia8	8.316
collider	Delphes	3.5.1
collider	MadAnalysis5	1.11.0 (2025-04-23)
collider	LHAPDF	6.5.5
collider	HepMC	2.06.09
collider	ROOT	6.28/10
collider	Python	3.10.12; numpy 1.26.4, scipy 1.15.3, awkward 2.9.0, matplotlib 3.10.8
mma-het	Wolfram Engine	13.3.0
mma-het	FeynRules	2.3.49 (29 Sep 2021)
micromegas	micrOmegas / CalcHEP	6.3.0 / 3.9.2
agent runtime	Python analysis stack	Python 3.13.12; uproot 5.7.4, awkward 2.9.0, numpy 2.4.4, scipy 1.15.3
agent runtime	Claude Code	2.1.273 (re-runs of 2026-09-16); original runs: see note
agent runtime	LLM	Claude Opus 4.6 (claude-opus-4-6), still served on 2026-09-16

Proposed replacement rows (booktabs, same layout as the current table):

```
collider    & MadGraph5_aMC@NLO & 3.7.0 \\
            & Pythia8      & 8.316 \\
            & Delphes    & 3.5.1 \\
            & MadAnalysis5 & 1.11.0 \\
            & LHAPDF     & 6.5.5 \\
```

```

& HepMC                & 2.06.09 \\
& ROOT                  & 6.28/10 \\
& pyhf                  & 0.7.6 \\
mma-het & Wolfram Engine    & 13.3.0 \\
& FeynRules             & 2.3.49 \\
agent runtime & Claude Code & 2.1.x (AUTHORS: version of the original runs from the usage logs
the 2026-09-16 re-runs used 2.1.273) \\
& LLM                   & Claude Opus 4.6 \\
& uproot, awkward, numpy, scipy & 5.7.4, 2.9.0, 2.4.4, 1.15.3 \\

```

Note on the Claude Code version: no session log from the April–May runs exists on this machine (the earliest local logs are from August 2026), so the original version has to come from the co-author who ran them (**AUTHOR**). The rows above are otherwise complete.

2 sm.pdf p.5, “LLM inference accounts for a minor fraction [AUTHORS: fill in numbers ...]”

Status: **PARTIAL** — measured for the ALP EFT benchmark with three current models; the Opus 4.6 numbers for all eight benchmarks are being produced now. The claim as written is *not* supported for the light benchmarks: attributing every gap before a model message to inference and every gap before a tool result to tool execution (main session plus stage subagents), the model share of wall-clock was 38% (Opus 5), 54% (Opus 4.8) and 51% (Sonnet 5); time inside `magnus run/status` calls was 60%, 45% and 47% (the two overlap when a stage polls a running job). Proposed wording (with the 21 Opus 4.6 runs of Table S3, model share 0.23–0.80, median 0.55):

For the light benchmarks of Table S3 the wall-clock time splits roughly evenly between LLM inference and cluster jobs (model share 0.2–0.8, median 0.55); for the Dark-SMEFT campaigns, where each campaign submits more than 100 jobs, cluster time dominates.

3 Table S3 (sm.pdf p. 6) — resource usage; also paper.pdf p. 5 and p. 7, reply.pdf p. 4 and p. 6

Status: **PARTIAL** — Claude Opus 4.6 (effort **xhigh**, v2 skills, cold memory), three attempts per benchmark for the seven figure benchmarks (attempt 1 launched 2026-09-16 22:00 UTC, attempts 2–3 on 2026-09-24). The two Delphes-level benchmarks have one attempt each and stay TBD (footnotes a, b); their row shows that single attempt. Values are means over the successful attempts (n = successful/attempted; per-attempt values in `docs/paper/results-2026-09-24/runs.csv`, figures in `.../figures/`). Tokens = main session + all sub-agents (uncached + cache-write + cache-read); files = distinct paths written or edited; Magnus jobs = `magnus run` submissions; USD from the CLI’s `total_cost_usd`.

Benchmark	n	Wall [h]	Sub-agents	Magnus jobs	Files	Tokens in [M]	Tokens out [k]	USD	Result
Scalar LQ, m_{ej} [Fig. 2(a)]	0/1	6.41	2	42	28	214.0	220	–	TBD ^a
ALP EFT, $\not{E}_T$ [Fig. 2(b)]	3/3	1.01	3.0	6.7	16.7	15.1	31	12.8	success*
U(1)’ scan [Fig. 2(c)]	3/3	0.92	3.3	17.0	22.3	4.4	30	7.6	success*
U1 LQ mono- τ [Fig. 2(d)]	0/1	4.00	2	29	19	181.4	219	106	TBD ^b
Heavy N [Fig. S2(a,b)]	3/3	0.60	3.0	9.0	14.3	3.9	17	4.8	success*
General Z' [Fig. S2(c)]	3/3	0.73	3.0	7.3	14.3	8.3	25	7.2	success*
KK graviton [Fig. S2(d)]	3/3	0.95	2.7	9.3	14.3	3.8	28	6.4	success*
U1 LQ at MuC, η [Fig. S2(e)]	2/3	0.68	3.0	14.0	6.0	4.4	18	7.8	success*, 1 failure ^c
U1 LQ at MuC, reach [Fig. S2(f)]	3/3	1.22	3.0	19.0	21.7	13.3	39	12.7	success*

* end-to-end reproduction with a quantitative deviation recorded as a footnote in at least one attempt (texts in `docs/paper/results-2026-09-24/tables.md`). Ranges over the 21 attempts of the seven figure benchmarks: 0.37–1.41 h, 3.6–22.2 USD, 1.3–36.9 M input tokens.

^a **TBD (handed over)**. Run stopped by the operator after 6.4 h without a figure. Model, validation and the 100 000-event LUXlep parton sample (lepton→photon LHE workaround) were done; two Pythia8+Delphes launches of the full sample finished on the cluster but failed at upload with “No space left on device” (the Delphes ROOT file alone is 27 GB against the job’s 10 GB packing space); a third launch that deletes HepMC/ROOT after the run (MG5 ! shell lines after the final **done**, keeping the LHCO) was running when stopped. The v2 skills now document the limit and the cleanup (commit 61800a9); a rerun is expected to take 1.5–2 h.

^b **TBD (handed over)**. Attempt 1 generated all 18 Delphes runs (4.0 h, 106 USD; four 9-run launches were lost to the same disk limit before the agent split them), then the orchestrator ended its turn with “waiting for event file downloads” and the headless session terminated without the analysis stage (failure mode (iv) in the paper’s classification). A continuation session in the same workspace produced the figure in 1.7 h / 29.5 USD with the reference’s shape but a 20–35 % weaker exclusion. The orchestrator skill now forbids ending a turn to wait (commit e7c5b61).

^c Attempt 2 of U1 LQ at MuC, η : both leptoquark curves coincide with the SM background to 2–3 % per bin, i.e. the new-physics contribution did not enter event generation (failure mode (i), *model*); attempts 1 and 3 reproduce the reference (attempt 1 without triggering the orchestrator: the main session ran the pipeline itself, 0 sub-agent calls).

Model-latency share of wall-clock (transcript timestamps) over the 21 attempts of the figure benchmarks: 0.23–0.80, median 0.55; the sentence “LLM inference accounts for a minor fraction” (sm.pdf p. 5) should be replaced by, e.g.:

For the figure benchmarks of Table S3 the wall-clock time is shared about equally between LLM inference and cluster jobs (model share 0.2–0.8, median 0.55 over 21 runs); cluster time dominates only for the Delphes-level benchmarks and the Dark-SMEFT campaigns.

4 Tables S4 and S5 — state on 2026-09-24

Table S5 (robustness, Opus 4.6 $\times 3$) is complete for the seven figure benchmarks: 20 of 21 attempts succeeded (the one failure is footnote *c* above). The two Delphes-level benchmarks have one attempt each (TBD^a, failure^b) and still need their $\times 3$ (1.5–6 h and up to >100 USD each at **xhigh**; use `--effort high`).

Table S4: column A = Opus 4.6 with Claude Code; column B = GPT-5.5 with Codex CLI 0.155.1 on branch `codex-com` (skills-only mode, same prompts, one clean directory per attempt, same judge, metrics from the Codex event stream and the main/sub-agent thread rollouts). GPT-5.3-Codex, the model contemporaneous with Opus 4.6, is not served to a ChatGPT-login Codex account (an API key is required), so GPT-5.5 was used; column C is left blank. Codex runs on a subscription, so no cost is recorded for column B.

Benchmark	Table S4 columns			single runs, 2026-09-16, default effort		
	A: Opus 4.6 ($\times 3$)	B: GPT-5.5 / Codex	C	Opus 4.8	Opus 5	Sonnet 5
Scalar LQ	0/1 ^a	1/1 ^{*e} (4.41 h, 24 jobs, 75.0 M tok)	–	–	–	–
ALP EFT	3/3	1/1 (0.99 h, 9 jobs, 13.4 M tok)	–	1/1	1/1	1/1
U(1)’ scan	3/3	1/1 (1.03 h, 21 jobs, 11.1 M tok)	–	–	1/1 [*]	–
U1 LQ mono- τ	0/1 ^b	1/1 ^{*d} (2.55 h, 20 jobs, 49.4 M tok)	–	–	–	–
Heavy N	3/3	–	–	–	–	–
General Z’	3/3	–	–	–	–	–
KK graviton	3/3	–	–	–	–	–
U1 LQ at MuC, η	2/3 ^c	–	–	–	–	–
U1 LQ at MuC, reach	3/3	–	–	–	–	–

^d Codex/GPT-5.5 completed mono- τ end to end (18 Delphes samples, 180 000 events): the $R_{D^{(*)}}$ bands match the reference, the exclusion boundary has the reference’s shape but is about 30 % weaker ($\sqrt{|g_c g_b|} \approx 1.07$ at 0.8 TeV against ≈ 0.8). The Opus 4.6 continuation run of attempt 1 was 20–35 % weaker in the same way, so two independent implementations agree with each other and not with the paper; the deviation is more likely a property of the analysis procedure the prompt prescribes (the LHC0 output carries no separate τ collection; both runs used type-4 jets as hadronic- τ proxies) than an agent error.

^e Codex/GPT-5.5 completed Scalar LQ end to end in 4.4 h: 100 000 LUXlep parton events, lepton $\rightarrow$ photon LHE substitution, Pythia8+Delphes (ATLAS card) to LHC0 after two failed detector passes (the interactive re-shower could not materialise `pythia8_card.dat`; “No space left on device”), the third as a non-interactive launch. The m_{ej} signal peaks at 3 TeV with ≈ 1.5 events/bin/100 fb $^{-1}$ against ≈ 3 in the reference (factor 2 low) and ends at 3.8 TeV; the Delphes card used anti- k_T $R = 0.6$ instead of 0.4. Opus 4.6 attempt 1 was stopped at the same stage after 6.4 h, so the benchmark is completable with the current skills and its Opus 4.6 $\times 3$ is worth running.

Correction to the U(1)’ note of 2026-09-17: the Codex/GPT-5.5 run reproduces the paper’s 3 TeV contour tip ($\tilde{g} \approx -0.72$, $g'_1 \approx 0.65$), whereas all three Opus 4.6 attempts and the Opus 5 run give $g'_1 \approx 0.48$. The 25 % shortfall is therefore an analysis choice made in the Claude runs, not a limitation of the prompt; the earlier wording “systematic to the prompt’s analysis procedure” is withdrawn (it stays a footnote, the runs stay successful under the agreed criterion).

Success criterion used (agreed 2026-09-16): the run completes end to end and the figure agrees with the requested part of the reference qualitatively; quantitative deviations go to a footnote (*). Failure modes follow the paper’s four classes; every verdict was produced by the assistive judge (`scripts/bench/judge.sh`) and confirmed by eye.

Still open:

1. Scalar LQ and mono- τ , Opus 4.6 $\times 3$: `scripts/bench/run_benchmark.sh 2005.06475 2 claude-opus-4-6 --effort high` and ... 1811.07920 3 ...; both root causes are fixed in the skills.
2. Column B with GPT-5.3-Codex: `codex login --with-api-key`, then `scripts/bench/run_benchmark_codex.<arxiv> <fig> gpt-5.3-codex`; the remaining five figure benchmarks and the $\times 3$ for column B (about 1 h each).
3. Column C (Sonnet 5 or Opus 5): `scripts/bench/run_benchmark.sh <arxiv> <fig> claude-sonnet-5`.

5 Other markers

Where	Marker	Status	Proposed action
paper.pdf p. 5	“skills-only mode executes the hello-world workflow of Sec. S-III [confirm that this has been tested]” (Codex/Cursor)	AUTHOR	The Codex-compatible layout exists (branch <code>codex-com: .agents/skills, .codex/agents, AGENTS.md</code>). Run the hello-world prompt with Codex on that branch once and keep the sentence; otherwise weaken to “is designed to”.
paper.pdf p. 7, Table I caption	“tokens consumed and wall-clock time [fill in]”	RUNS	Same numbers as Table S3 (Opus 4.6 column); cite Table S3 instead of repeating them.
paper.pdf p. 9–10	“[confirm the extent of human intervention between campaigns]”	AUTHOR	Dark-SMEFT campaigns were not re-run; statement must come from the campaign logs.

Where	Marker	Status	Proposed action
paper.pdf p. 10; reply.pdf p. 6	“several weeks of work for an experienced phenomenologist [adjust estimate]”	AUTHOR	Judgement call; unchanged by measurements.
reply.pdf p. 1	checklist (1)–(4): fill S3/S4/S5, confirm human intervention, set <code>\markupfalse</code> , remove red markers	PARTIAL	(1) in progress, (2) author, (3)–(4) mechanical once (1)–(2) are done.
reply.pdf p. 4	“the LLM token consumption [fill in from the usage logs]”	RUNS	From Table S3; note the main+sub-agent definition.
reply.pdf p. 4	“The wall-clock time of a benchmark (1–6 hours) ...”	DONE	Consistent with today’s runs (0.9–1.4 h for the lightest benchmark); keep.
sm.pdf p. 4	“archived release ... DOI and per-run provenance packaged as Workflow Run RO-Crates [confirm or remove]”	AUTHOR	No RO-Crate export exists in the repository today; either remove or keep as a plan.
sm.pdf p. 5	Table S4 placeholder text “e.g. Claude Opus 4.6 vs. Claude Sonnet, GPT-5.x via Codex, Gemini via ADK”	PARTIAL	Replace with the three columns chosen above; drop Gemini/ADK unless the python-agent path is run.
sm.pdf p. 6	Table S3 Dark-SMEFT rows (sub-agent calls, tokens)	AUTHOR	Not reproducible here; from the campaign logs.

6 Provenance of the numbers

Branch `feat/skill-evolution-v2` of the ColliderAgent repository. All benchmark data are in `docs/paper/results-tables.md` (Tables S3–S5 and every footnote text, generated by `scripts/bench/aggregate.py`), `runs.csv` (one row per attempt), `documented_runs.json` (the 2026-09-16 attempts, whose sandboxes were deleted; numbers copied from `docs/paper-tbd-audit-2026-09-16.md`) and `figures/` (the produced figure of every attempt). The sandboxes under `bench_runs/` were deleted after aggregation. Image versions come from the probe blueprint `collider-env-probe` on zhustation. Skills used are the v2 skills (context-lean rewrite of 2026-09-16), so the Opus 4.6 numbers measure the current system.